FIREWALL and ROUTING - Detailed Notes

What are Firewalls and Routing?

Let's start with a simple real-world example:

Imagine your computer is like a house:

- A firewall is like a security guard at your door. It checks who's trying to get in or out and either lets them in or blocks them.
- Routing is like a GPS system that tells your data where to go across the internet or networks.

What is a Firewall?

Definition:

A firewall is a security system (hardware or software) that controls the traffic (data) going in and out of a network or device.

Purpose:

- Protect your device/network from hackers, malware, and unauthorized access.
- Control what websites or services can be accessed.
- Monitor and log traffic for security checks.

Types of Firewalls:

Type	What It Does (Simple)
Packet-filtering	Checks simple info (IP address, port) to allow or block data
Stateful	Remembers past traffic to help with decisions
Application-level	Understands apps like Facebook or YouTube to filter them

Next-Gen (NGFW)	Advanced: combines firewall + antivirus + malware scanning
Host-based	Runs on your PC (e.g. Windows Firewall)
Cloud Firewall	Used in cloud services like AWS or Azure

How Firewalls Work (Simple Steps):

1. Data packet arrives at firewall.
2. Firewall checks its rules (like a checklist).
3. If packet matches a rule, it is allowed or blocked.
4. If no rule matches, it's denied (default-deny rule).

Real-Life Example:

You have a home Wi-Fi router:

- You configure the firewall to block gaming websites after 9 PM for your child.
- It also blocks hackers from scanning your smart TV or laptop.

Stateful vs Stateless Firewalls

Feature	Stateless	Stateful
Remembers traffic?	No	Yes (remembers ongoing connections)
More secure?	Less secure	More secure
Example	Basic firewall	Modern firewalls

What is NAT (Network Address Translation)?

- NAT lets many devices use one public IP address to access the internet.
- Your home Wi-Fi router uses NAT to connect all your devices to the internet.
- NAT hides internal IPs from outsiders – adds a layer of security.

Common Firewall Uses:

- Block or allow websites, ports, or apps.
- Create rules for VPN access.
- Set up a DMZ (special zone for public-facing servers).
- Limit what devices can talk to each other.

Basic Firewall Best Practices:

- Default Deny: Block everything first, allow only what's needed.
- Update regularly: Always keep firmware/software updated.
- Log and monitor: Know what's happening in your network.
- Separate zones: Keep guest network separate from main network.
- Backup your settings: Always have a copy of your config.

Firewall Evolution

Generation	Name	Key Feature	Time Period
1st	Packet Filtering	Filters by IP/Port	Late 1980s
2nd	Stateful Inspection	Tracks active connections	Early 1990s
3rd	Application Proxy	Understands specific protocols (HTTP etc.)	Mid 1990s

4th	UTM	All-in-one security box	2000s
5th	Next-Gen Firewall (NGFW)	Application, user, and content control	2010s
6th	Cloud / AI Firewalls	AI, cloud-native, firewall-as-a-service	2015 – Today

What is Routing?

Definition:

Routing is the process of choosing the best path for your data to travel across networks.

Think of it like sending a letter:

- You need the correct address.
- The postal system (router) decides the best route to deliver it.

What is a Router?

- A router is a device that connects two or more networks together.
- It reads the destination IP address and decides which direction to send the data.

Routing Types:

Type	Simple Meaning	Example
Static Routing	Routes you set manually	Small home network
Dynamic Routing	Router learns paths automatically	Large business or ISP network

Common Dynamic Routing Protocols:

Protocol	Used for	Easy Notes
RIP	Small networks (old)	Simple but slow and limited
OSPF	Medium to large networks	Fast and popular in enterprises
BGP	Internet routing between ISPs	Used on the internet (very advanced)

How Routers Work (Steps):

1. Your computer wants to send data to Google.
2. It sends data to the local router (default gateway).
3. The router checks its routing table.
4. It forwards the data to the next hop.
5. This continues until it reaches Google's server.

How Firewalls and Routing Work Together

Feature	Firewall	Router
Controls access	Yes	No (not usually)
Chooses path	No	Yes
Can be combined?	Yes, many routers include firewalls	